TREDENCE AI GRADS · PROJECT EVALUATION

Solution Core50

Milestone 1 · Healthcare Liver Disease
Submission:
Submission-Milestone1/M1_Healthcare_Solution_Core50.ipynb
Evaluated: 06 Nov 2026

90 / 100
Distinction · 90%

Criterion-wise marks

Data Understanding & Quality Checks — 18 / 20

Strong data understanding and quality checks are demonstrated, including target verification and missingness inspection. However, there is no explicit mention or handling of lookup tables (Cityname_statecode.xlsx, Statecode_name.xlsx) or other supplied files, which could have been inspected or joined if relevant. Including these would improve completeness.

Exploratory Data Analysis (EDA) — 18 / 20

The EDA covers target balance and clinical feature distributions well with appropriate plots. However, there is no deeper statistical summary or exploration of relationships between features or with the target beyond univariate distributions. Adding correlation analysis or bivariate plots would strengthen the EDA.

Data Cleaning & Feature Engineering — 17 / 20

Data cleaning is handled transparently with leakage-aware preprocessing pipelines. The exclusion of IDs and imputation strategies are appropriate. However, the submission does not discuss or handle outliers or engineer new features, which were part of the milestone requirements.

Linear Model Building — 20 / 20

The baseline logistic regression model is well implemented with appropriate stratified splitting, preprocessing, and class imbalance handling. The pipeline ensures no data leakage.

Evaluation & Findings — 17 / 20

Evaluation metrics are comprehensive and appropriate. The discussion of false negatives and limitations is present but brief. More detailed interpretation of results and implications would improve educational value.

Strengths

• Thorough data loading, target verification, and quality checks with missingness and duplicates reported.
• Clear and appropriate EDA plots for target and clinical features.
• Leakage-aware preprocessing pipeline with imputation, scaling, and encoding.
• Well-implemented logistic regression baseline with stratified splitting and class imbalance handling.
• Comprehensive evaluation metrics reported with responsible interpretation and safety disclaimers.

To improve next time

• Include inspection or integration of all supplied lookup tables and files to confirm identifiers and enrich data understanding.
• Expand EDA to include bivariate analyses, correlations, or feature-target relationships.
• Address outliers explicitly and perform feature engineering beyond basic encoding and imputation.
• Provide deeper interpretation of model results, false negatives, and educational insights.

Overall feedback

The submission meets the core requirements of Milestone 1 with a clear, reproducible workflow covering data understanding, EDA, preprocessing, baseline logistic regression modeling, and evaluation. It responsibly handles data leakage and class imbalance and reports appropriate metrics with safety considerations. To advance beyond the core level, the work should incorporate more comprehensive data integration, richer exploratory analyses, explicit outlier handling, feature engineering, and deeper interpretation of findings. Overall, this is a solid foundational submission suitable for educational purposes but could be enhanced with more detailed analysis and integration of all provided data resources.

AI-generated evaluation against the milestone rubric. Subject to trainer review. Each submission is evaluated once; this score is recorded.